

Project Demo Planning

Date	03 July 2026
Team ID	N/A
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Project Demo Planning:

This document outlines the plan for demonstrating OptiCrop, including the flow of the demo, key features to highlight, and timing for each section.

S. No	Demo Section	Description	Duration (mins)	Responsible Member
1	Introduction & Problem Statement	Explain the agricultural problem — farmers struggling to choose the right crop for their soil/climate — and OptiCrop's goal	2	Vandrase Durga Arun kumar
2	Dataset & EDA Walkthrough	Show the crop dataset and key visualizations (feature distributions, humidity vs. crop, correlation heatmap)	3	Vandrase Durga Arun kumar
3	Model Building & Comparison	Walk through training KNN, Logistic Regression, Decision Tree, and Random Forest, and comparing their accuracy/precision/recall/F1	3	Vandrase Durga Arun kumar
4	Live Application Demo	Open the Flask app, enter sample N/P/K/temperature/humidity/pH/rainfall values, and get a live crop recommendation	4	Vandrase Durga Arun kumar
5	Code & Architecture Walkthrough	Briefly show the project structure (data/, model/, templates/, app.py) and how the model is loaded and used	2	Vandrase Durga Arun kumar
6	Scalability & Future Plan + Q&A;	Summarize current limitations, the future roadmap, and open the floor for questions	3	Vandrase Durga Arun kumar

Demo Flow Summary:

Step	Activity	Notes
1	Introduction & Problem Statement	Keep it brief — set up why crop selection matters before diving into the solution
2	Solution Overview	Cover the ML approach (multi-algorithm comparison) and the Flask app architecture
3	Live Feature Demonstration	Run at least 2–3 different input combinations to show varied, correct predictions

Step	Activity	Notes
4	Q&A; Session	Be ready to discuss model choice (why Random Forest), data source, and scalability plans